

SUSAN T. PARKER, PhD MPP

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CURRENT POSITION

2024-present Research Assistant Professor, Department of Emergency Medicine, Feinberg School of Medicine, Northwestern University

2024-present Member, Buehler Center for Health Policy and Economics

EDUCATION

2023 Ph.D., Health Policy (Specialization: Economics), University of Michigan, Ann Arbor, MI

2017 MS, Analytics (Applied Data Science), University of Chicago, Chicago, IL

2011 MPP, Public Policy, University of Chicago, Chicago, IL

2009 BA, International Relations, Michigan State University, East Lansing, MI

PREVIOUS EMPLOYMENT

2023-24 Postdoctoral Research Fellow, Community to Community (C2C) Public Policy Initiative, School of Public Policy and Urban Affairs & Bouvé College of Health Sciences, Northeastern University

2016-18 Senior Data Scientist, CivicScape, Chicago, IL

2015-16 Data Scientist, Midwest Physician Administrative Services, Downers Grove, IL

2013-15 Research Manager, University of Chicago Crime Lab, Chicago, IL

2011-13 Analyst, Obama for America, Chicago, IL

PUBLICATIONS: Peer-Reviewed

2025 Parker, S.T. "Supervised Natural Language Processing Classification of Violent Death Narratives: Development and Assessment of a Compact Large Language Model." *JMIR AI*. <https://doi.org/10.2196/68212>.

2024 Parker, S.T. "The Association Between Concealed Handgun License Laws and Concealed Handgun License Adoption, 2002-2019." *American Journal of Preventive Medicine*. <https://doi.org/10.1016/j.amepre.2024.07.009>.

2024 Gilbert, L., Parker, S.T., Schechter, L. "The Effects of Lockdown Orders on Domestic Violence: Evidence from Medical Records." *Review of Economics of the Household*. <https://doi.org/10.1007/s11150-024-09696-8>

2022 Barber, C., Cook, P.J., & Parker, S.T. "The emerging infrastructure of US firearms injury data." *Preventive Medicine*. <https://doi.org/10.1016/j.ypmed.2022.107129>

2022 Parker, S.T. "Measuring gun violence in police data sources: transitioning to NIBRS." *Injury Epidemiology*, 9(1). <https://doi.org/10.1186/s40621-022-00376-8>

- 2020 Parker, S.T. “Estimating Nonfatal Gunshot Injury Locations With Natural Language Processing and Machine Learning Models.” *JAMA Network Open*, 3(10). <https://doi.org/10.1001/jamanetworkopen.2020.20664>
- 2017 Collins, M, Parker, S.T., Scott, T. L., Wellford C.F. “A Comparative Analysis of Crime Guns: Implications for Reducing Violence.” *RSF: The Russell Sage Foundation Journal of the Social Sciences*, 3(5). <https://doi.org/10.7758/rsf.2017.3.5.05>
- 2015 Cook, P.J., Parker, S.T., Pollack, H.A. “Sources of guns to dangerous people: what we learn by asking them.” *Preventive Medicine*, 79. <https://doi.org/10.1016/j.ypmed.2015.04.021>

PUBLICATIONS: Not Peer Reviewed

- 2025 Azrael, D., Parker, S.T. “Trends in Stolen Firearms from 2018 to 2022: Leveraging Data for Informed Crime Policy.” Council on Criminal Justice. <https://counciloncj.org/trends-in-gun-theft/>
- 2025 Barber, C., Cook, P. J., Parker, S.T. “Nonfatal Firearm Injury Surveillance: An Update.” *Submitted to National Institute of Statistical Science*
- 2022 Cook, P. J., Parker, S.T. Correcting Misinformation on Firearm Injury Intent. *JAMA Network Open*. 5(12). <https://doi:10.1001/jamanetworkopen.2022.46434>
- 2022 Cook, P. J., Parker, S.T. Reducing Youth Firearm Carrying: Focus on the Parents. *JAMA Network Open*. 5(4). <https://doi.org/10.1001/jamanetworkopen.2022.5136>
- 2021 Parker, S.T. “Measuring Gun Violence Using Police Data.” in Roman, John K and Philip J. Cook, eds. *Improving Data Infrastructure to Reduce Firearms Violence*. Chicago, IL: NORC at the University of Chicago, 2021.

Working Papers

- Gangaram, A., Parker, S.T., “The Effects of School Shootings on Firearm Behaviors.” *Under Review*.
- Parker, S.T. , Ross, M.B., Ross, S.L. “The Impact of Prohibiting Pretextual Enforcement on Traffic Stops, Accidents, and Crime.” *NBER Working Paper. Under Review*.
- Parker, S.T. “Jumping the Gun: The Effects of Permitless Carry Laws on Crime.” *Under Review*.
- Parker, S.T., Fischer, S., Miller, M., Azrael, D. “Identifying Suicides that Occur in the Presence of Police Using Natural Language Processing (NLP).” *Preparing for Submission. Draft Available*.
- Magee, L.A., Ortego, A., Parker, S.T., Dycus, D., Adams, K. “Victim Assistance Units and Solving Nonfatal Shootings.” *Preparing for Submission. Draft Available*.
- Gong, X., Han, Y., Parker, S.T., Ross, M.B., Ross, S.L. Noise or Signal? Decomposing the Sources of Officer Disparity in Traffic Stops. *Preparing for Submission. Draft Available*.

In Progress

- Gong, X., Han, Y., Parker, S.T., Ross, M.B., Ross, S.L., “Testing for Discrimination in Police Traffic Stops using Telemetric Mobility Data.” *Preparing Manuscript*.
- Parker, S.T. “Do Permitless Carry Laws Reduce Racial Disparities in Policing?” *Preparing Manuscript*.
- Lundberg, A.L., Parker, S.T. “The Criminal Justice and Economic Effects of the Illinois SAFE-T Act.” *Preparing Manuscript*.

Parker, S.T., Kim, P., Huang S., Welch, S., Beidas, R., Mason, M. “Predicting intervention uptake: Modeling drug overdose harm reduction training outreach effectiveness.” *Preparing Manuscript.*

Parker, S.T., Richard, K., Schechter, L. “Family Violence Effects of Social Safety Net Incentives for Child Support Cooperation.”

Magee, L.A., Parker, S T.. “Identifying Barriers to Solving Nonfatal Shootings: A Large Language Model Approach.”

AWARDS AND FELLOWSHIPS

2025	Northwestern University Clinical and Translational Science (NUCATS) Pilot Grant Award (\$50,000)
2024	National Violent Death Reporting System (NVDRS) New Investigator Award (\$7,000)
2022	Rackham Graduate Student Conference Travel Grant (\$900)
2022	Martha and Ernest Hammel Graduate Student Research Award, joint with Anjelica Gangaram (\$3,100)
2021	Microsoft Computational Justice Award (\$20,000 computing credit)
2020	Walter J. McNerney Award - COVID-19 Rapid Small Grant Award (\$10,000)
2019	Rackham Pre-Doctoral Research Award (\$1,500)
2019	Firearm Safety Among Children and Teens (FACTS) Student Poster Award
2015	Capstone thesis award, University of Chicago Analytics Program
2009	Phi Beta Kappa
2005-09	Skandalaris Family Foundation Scholarship

SPONSORED RESEARCH

Title: Criminal Justice-based Advocate Intervention Randomized Control Trial (\$377,400)

Role: Co-Investigator (PI: Lauren A. Magee)

Source: Arnold Ventures

Dates: 09/01/2025-08/31/2029

Title: Do Lethality Assessment Programs Improve Domestic Violence Outcomes? (\$64,000)

Role: Principal Investigator

Source: Arnold Ventures - Planning Grant

Dates: 09/01/2025-08/31/2026

Title: Reducing Barriers to Developing and Applying Accurate and Fair Natural Language Processing (NLP) Methods to the National Violent Death Reporting System (\$175,000)

Role: Co-Investigator

Source: Centers for Disease Control (CDC) / American Public Health Association (APHA) Data Science Demonstration Project Grant

Dates: 04/01/2023-04/01/2024

Title: The Effect of School Shootings on Firearm Behaviors (\$17,000)

Role: Co-Principal Investigator

Source: University of Michigan School of Public Health / Public Health IDEAS for Preventing Firearm Injuries Initiative

Dates: 06/01/2022-01/31/2023

Title: The Effect of Pandemic Shutdown Orders on Domestic Violence (\$33,000)

Role: Co-Principal Investigator

Source: Gates Foundation - Covid 19 Research Database Research Accelerator Program

Dates: 08/01/2021-08/01/2022

Title: The Effect of Concealed Carry Laws on Firearm Injury

Role: Project Director, Grant Author

Source: National Gun Collaborative for Gun Violence Research (NCGVR) (\$235,000)

Dates: 07/15/2019-12/31/2021

CONFERENCES AND SEMINARS

- 2025 National Research Conference for Prevention of Firearm-Related Harms - 11/2025, NBER Economics of Firearms Markets - 10/2025, Northwestern University Clinical and Translational Science (NUCATS) Institute - 10/2025, NORC Innovation Day - 06/2025, ASHEcon Discussant, Organizer - 06/2025
- 2024 The Center for Neighborhood Engaged Research & Science (CORNERS) - Northwestern University - 12/2024, National Research Conference for Prevention of Firearm-Related Harms - 12/2024, APPAM - 11/2024, University of Indiana - 10/2024, APHA / CDC Data Science Demonstration Conference - 8/2024, ASHEcon Presenter, Organizer - 6/2024, Council on Criminal Justice Crime Trends Working Group - 04/2024, National Institute for Statistical Science (NISS) 2nd IOF Workshop on Gun Violence - 5/2024. Johns Hopkins Bloomberg School of Public Health - 3/2024.
- 2023 APPAM - 11/2023, National Research Conference for Prevention of Firearm-Related Harms - 11/2023. ASHEcon Presenter, Organizer - 6/2023, University of Massachusetts Boston Department of Urban Public Health - 2/2023, University of Iowa School of Public Health (Online) - 2/2023, Georgia Tech School of Public Policy - 2/2023.
- 2022 State University of New York (SUNY)-Albany School of Criminal Justice - 12/2022, National Firearm Injury Conference - 11/2022, Rutgers-Camden Department of Health Sciences (Online) - 11/2022, American Public Health Association (APHA) - 11/2022.
- 2021 FACTS Symposium (Online - 11/2022, Data Foundation, NORC, NPSC Webinar - 10/2022, Law Enforcement and Public Health (Online) - 3/2022.
- 2020 American Society of Criminology. Canceled due to COVID-19. FACTS Annual Symposium - 09/2020.
- 2015 American Society of Criminology - 11/2015. International Association of Chiefs of Police - 10/2015, Law and Society Association - 5/2015, American Public Health Association - 11/2015

TEACHING

Instructor - Introduction to SQL, Advanced SQL, Introduction to Python, Introduction to Machine Learning, 2016-2018

Graduate Student Courses

Graduate Teaching Assistant - Health Economics II, Fall 2019

Graduate Teaching Assistant - Health Economics I, Winter 2020

Graduate Teaching Assistant - Data Driven Methods for Public Health, Winter 2021

Graduate Teaching Assistant - Quantitative Methods for Social Science, Spring 2021

SERVICE

Journal reviewer: American Journal of Epidemiology, American Journal of Preventive Medicine, American Journal of Public Health, Health Economics, The Lancet - Child and Adolescent Health, JAMA Network-Open, Journal of Experimental Criminology, Journal of Crime and Justice, Journal of Medical Internet Research, Preventive Medicine

Internal Committee Member, University of Michigan Health Services Organization and Policy (HSOP) Admissions and Doctoral Committee (2022)

External Membership, Firearm Data Infrastructure Working Group, American Society of Health Economists (ASHEcon), Association for Public Policy Analysis and Management (APPAM)

Abstract Reviewer, National Research Conference for Prevention of Firearm-Related Harms

SKILLS

AWS, R, Python, SQL (PostgreSQL, PostGIS, Snowflake, Trino), STATA, Unix